Course Name: Computer Networks.
Course Code: CSA0702.

Slot : B.

Test : Short Answers.

Faculty Name: Dr. R. S. Selvan.

Assessment: Tool-1.

Total Mark : 20
Obtained mark: 18

Understanding of Concept	Explanation.	Application.	Organization.	Accuracy	clarity.
25%.	25%.	20%.	15%.	10%.	5%.
5	5	4	3	0.5	0.5

ASSESSMENT TOOL-1
CSA0702
COMPUTER NETWORKS.

Name: V. Sanjai
Reg No: 192312486.

1) Design File Transfer.

Given.

File size $= 150$ MB.
$= 150 \times 10^6$
$= 150,000,000$ bytes.
Segment size $= 1500$ bytes.
Frame overhead $= 40$ bytes
Bandwidth $= 50$ Mbps.

i) Number of segments.

$$\frac{150000000}{1500} = 100\,000 \text{ segments.}$$

ii) Number of frames.

Each segment becomes one frame.

Frames $= 100000$.

iii) Total Data Link overhead.

$= 100000 \times 40$
$= 4000000$ bytes.
$= 4$ MB.

iv) Total Transmitted data.

$= 150 + 4$
$= 154$ MB.

v) Transmission time.

Total bits ×8

$= 154 \times 8$

$= 1232$ Mbits.

$= \frac{1232}{50} = 24.64$ s.

② Sliding Window.

Given.

Window size = 6

Total frames = 48.

Windows required

$= \frac{48}{6} = 8$

Retransmissions.

1 frame lost in every window.

$= 8 \times 1$

$= 8$.

Flow control:

Flow control prevents sender overflow with receiver by matching transmission speed with receiver capacity. improving efficiency.

③ IP Fragmentation.

Given.

Packet = 12000 bytes.

MTU = 1500 bytes.

IP header = 20 bytes.

.. per fragment.

= 1500 - 20

= 1480 bytes.

fragments.

$= \left\lceil \frac{12000}{1480} \right\rceil$

= 9.

Header overhead.

= 9 × 20

= 180 bytes.

Network layer:

Uses fragmentation, routing, fragment offsets, identification fields, and reassembly at destination.

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(4) Sliding Window ~~[illegible]~~ ~~[illegible]~~ ~~[illegible]~~.

Given.

Window size - 6.

Total frames = 48.

Window required.

$= \frac{48}{6}$

= 8.

Retransmission:

1 frame lost in every window.

= 8 × 1

= 8.

Scanned with OKEN Scanner

5) Session Layer Synchronization:

File = 600 MB

checkpoint every 60 MB.

Failure after 390 MB.

checkpoints before failure:

$= \frac{390}{60}$

= 6 complete checkpoints.

Last checkpoint = 360 MB.

Retransmission.

= 390 − 360

= 30 MB.

Importance:

Synchronization allows communication to resume from the last checkpoint instead of restarting.

6) Presentation layer.

Original = 900 MB.

Compression = 40%.

Encryption overhead = 5%.

Compressed size

$= 900 \times 0.60$

$= 540$ MB.

Encrypted size.

$= 540 \times 1.05$

$= 564$ MB.

Percentage reduction.

$= \frac{900 - 564}{900} \times 100$

$= 34\%$.

Presentation layer.

Provides compression, encryption, decryption, and data format conversion.

(7) FTP Transfer:

Given.

Data $= 3$ GB

$= 3000$ MB.

~~Request size $= 3$ MB.~~

Throughput $= 100$ Mbps.

Requests.

$= \frac{3000}{3}$

$= 1000$

Transmission time.

Data in bits:

$= 3000 \times 8$

$= 24000$ Mbits.

$= \frac{24000}{120} = 200$ s.

Provides FTP services, authentication, file management and error handling.

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(8) End-to-End Delay:

Processing delay

Application $= 4$ ms.

Transport $= 3$ ms.

Network $= 5$ ms.

Data link $= 2$ ms

Physical $= 1$ ms.

Total processing:

$= 4 + 3 + 5 + 2 + 1$

$= 15$ ms.

Propagation:

18 ms.

Transmission:

12 ms.

Total delay:

= 15 + 18 + 12

= 45 ms

Layer contribution.

Application: User services.

Transport: Reliability.

Network: Routing.

Data link: Framing.

Physical: signal transmission.

Frame overhead:

Given.

Useful data = 80 MB

= 80000000 bytes

Frame = 1600 bytes.

Overhead = 80 bytes

Useful payload / frame:

= 1600 − 80

= 1520 bytes.

Frames.

$$= \left\lceil \frac{80000000}{1520} \right\rceil = 52,632$$

overhead

$= 52,632 \times 80$

$= 4210,560$ bytes.

Transmission efficiency.

$= \frac{1520}{1600} \times 100$

$= 95\%$.

The Higher protocol overhead reduces bandwidth available for useful data and lower efficiency.

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(15) Hospital Medical Record:

Original $= 240$ MB, Segment $= 1400$ bytes.
Compression $= 30\%$, Frame overhead $= 60$ bytes.

Compressed size:

$= 240 \times 0.70$

$= 168$ MB.

bytes:

168000000 bytes.

Segment:

$= \frac{168,000,000}{1400} = 120,000$.

overhead:

$= 120,000 \times 60$

$= 7200000$ bytes

$= 7.2$ MB.

Final transmitted data

$= 168 + 7.2$

$= 175.2$ MB

OSI Layer Cooperation:

Presentation: Compression and encryption.
Session: Synchronization and recovery.
Network: Routing and addressing.
Data Link: Framing and error detection.